

BLUE APRON

Social Engagement Analysis

Founded August 2012 — a fresh ingredient and recipe delivery service helping time-pressed, convenience-seeking customers cook healthy, delicious meals at home within a specified prep time.

This case applies the dataset to a Blue Apron-relevant marketing question; the source data does not name Blue Apron directly. Engagement rate is used as a proxy for awareness/reach, not a stand-in for revenue.

The Business Tension

1

Blue Apron competes on solving time pressure and decision fatigue. Marketing invests in influencer and content partnerships to reach that audience.

2

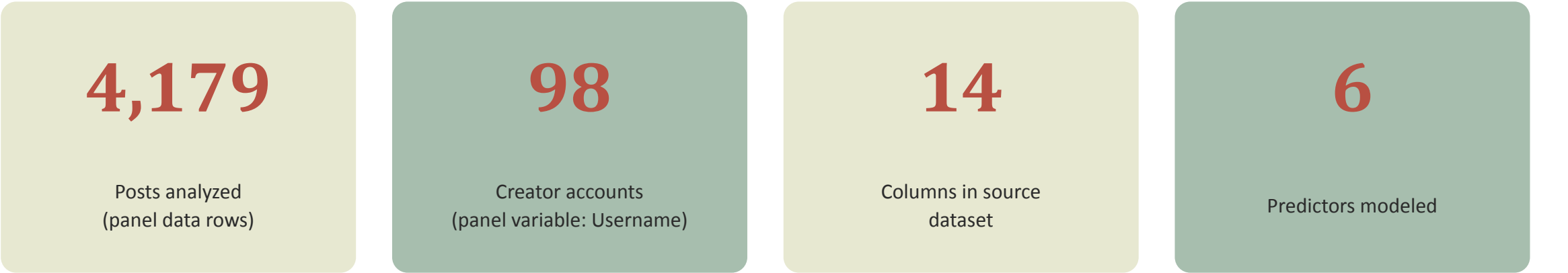
The default assumption: better content mechanics (hashtags, captions, photos) drive engagement and growth. This analysis tests that assumption directly.

OBJECTIVE

Determine whether engagement is driven by what is posted or by who posts it — and what that means for where marketing dollars should go.

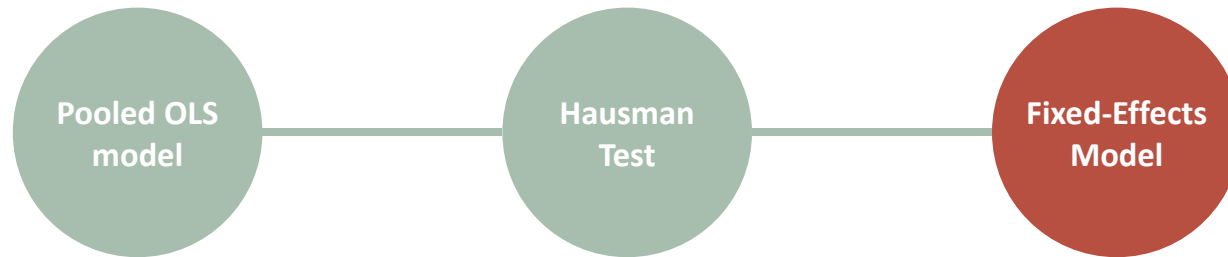
Engagement rate is treated as a leading indicator of awareness/reach efficiency in this analysis, not as a proxy for signups or revenue (see Limitations).

Data Overview



Panel variable	Username (98 distinct accounts)
Time / replication variable	PostID
Target variable	Engagement_rate (%)
Modeled predictors	Post_length, No. of hashtags, Photo (0/1), LikeCount, CommentCount, No. of mentions

Methodology: Why Fixed Effects



The random-effects model is preferred only if both RE and FE are appropriate. The Hausman test checks that assumption.

HAUSMAN TEST RESULT

Chi-sq

31.4

df

6

p-value

0.0000212

Only the FE model is consistent for this data.

Model Fit

187.3

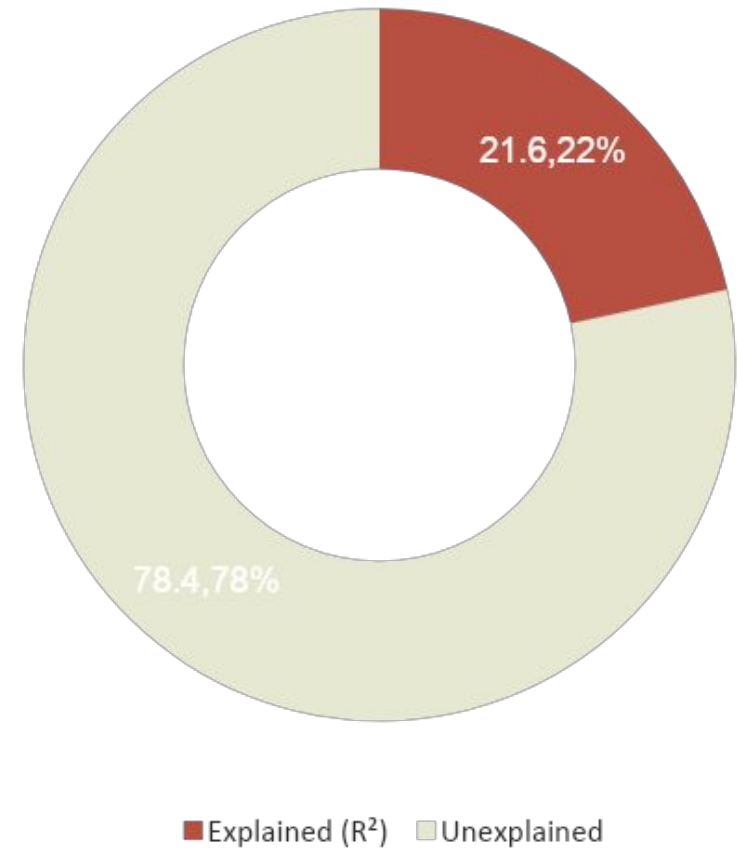
F-statistic (p-value = 0)

21.6%

R-squared

~78% of engagement-rate variation is driven by factors outside this model.

Explained vs. Unexplained Variation



Key Drivers: What Moves Engagement

LikeCount

SIGNIFICANT

Coefficient	0.00083093
t-value	30.28
p-value	< 0.0001

CommentCount

SIGNIFICANT

Coefficient	0.00526235
t-value	8.67
p-value	< 0.0001

The only two predictors statistically significant at the 5% level — engagement responds to likes and comments already accumulating on a post.

What Doesn't Move the Needle

Post_length

NOT SIGNIFICANT

Coefficient: -0.00008111

p-value: 0.6591

No. of hashtags

NOT SIGNIFICANT

Coefficient: -0.00340658

p-value: 0.7133

Photo presence

NOT SIGNIFICANT

Coefficient: 0.17679855

p-value: 0.2011

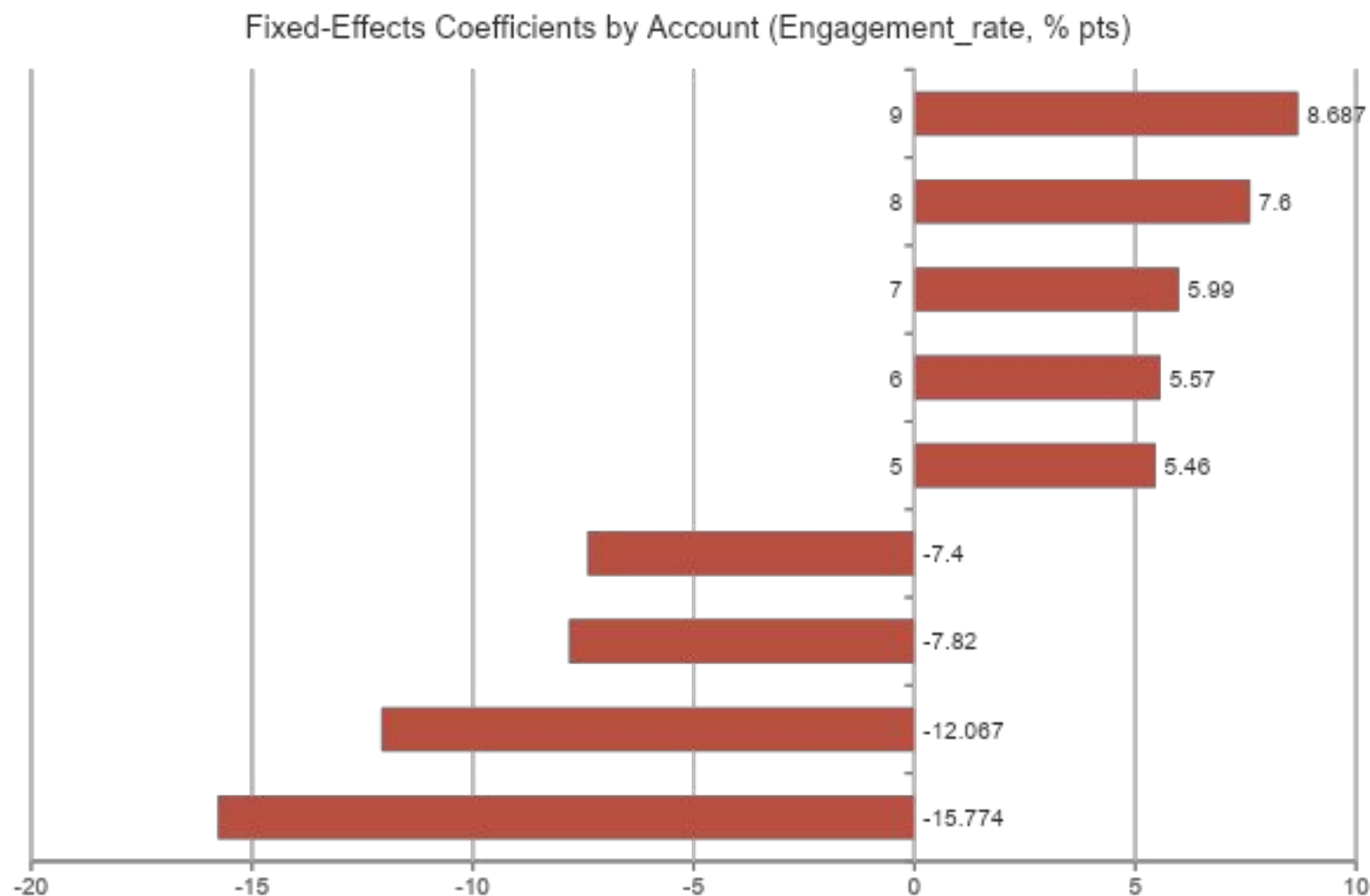
No. of mentions

NOT SIGNIFICANT

Coefficient: -0.02757196

p-value: 0.2180

Account Effects: Who Posts Matters



~24 pts

Spread from lowest to highest account fixed effect — larger than the effect of any single post-level variable.

Highest:

nikki_ferrara_nyc +8.69

Lowest:

the_hunger_diaries -15.77

→ see segmentation, next

Segmentation: Do Audiences Cluster by Niche?

Illustrative read from account names only — not verified creator-category data.

Top Performers

chefericsculinaryclassroom +2.41 *(Food/cooking)*

cook_ingwiththegreens +0.95 *(Food/cooking)*

winebarchronicles +1.05 *(Food/wine)*

aysoo_handmades +5.99 *(Handmade/crafts)*

milwaukeeemoms +2.82 *(Family/parenting)*

Bottom Performers

the_hunger_diaries -15.77 *(Food-themed name)*

travel.couples -2.29 *(Travel/lifestyle)*

missmustardseed -1.66 *(Home/DIY)*

dr.marielbuque -2.29 *(Unclear)*

overheardbumble -7.82 *(Unclear)*

No clean niche pattern emerges from names alone — food-themed accounts appear at both extremes (chefericsculinaryclassroom +2.41 vs. the_hunger_diaries -15.77). Verified creator-tier/category data (e.g., the dataset's unused "Nano" column) is needed before making a segmentation-based spend decision.

Correlation Diagnostics

Correlation shown is between the estimated coefficients (variance-covariance matrix) — not the raw post variables.

-0.339

Post_length
↔ No. of hashtags

-0.156

LikeCount
↔ CommentCount

+0.14

Post_length
↔ LikeCount

**-0.11
to
+0.07**

All other coefficient pairs

Limitations & Data Constraints

- 1 R-squared of 0.2162 leaves about 78% of engagement variation unexplained.
- 2 Engagement rate is a proxy for awareness/reach — this dataset has no conversion, signup, or revenue field to validate against.
- 3 Two extreme outlier posts (roughly 130-140% engagement rate) appear in the scatterplots, not flagged as treated.
- 4 Source dataset lists 14 columns; only 6 predictors plus panel/time variables are identified in this report.
- 5 Panel and heterogeneity charts display only 11 of 98 accounts; the fixed-effects chart caps at the top 50 accounts.
- 6 The source PDF does not explicitly reference Blue Apron — business framing was applied for this case exercise.

Recommendation: A Test-and-Learn Pilot

Author's recommendation — not stated in the source data

HYPOTHESIS

Creator/account identity — not post mechanics — drives engagement, so spend efficiency depends on WHO you partner with, not HOW they post.

TEST DESIGN

Reallocate a defined test tranche of upcoming influencer budget toward historically top-quartile creators (once tier/category is verified); hold a control group on the current mix.

SUCCESS METRIC

Track attributed signups or trial starts per dollar (via UTM/promo codes) — not engagement rate alone, since engagement is a proxy, not a revenue outcome.

DECISION RULE

If the test group shows a meaningfully higher signup conversion per dollar over one full content cycle, scale the reallocation; if not, revisit the hypothesis.

Measurement Roadmap & Next Steps

- 01** Add UTM/promo-code attribution so engagement can be linked to signups and revenue, not just reach.
- 02** Capture verified creator tier and content-category metadata (activate the dataset's unused "Nano" column; build a full niche taxonomy).
- 03** Code content themes and messaging (convenience, health, price, family) to test positioning — not just post mechanics.
- 04** Investigate and address outlier posts, and confirm the real-world tie to Blue Apron's influencer program, before scaling any spend decision.

Questions & Discussion